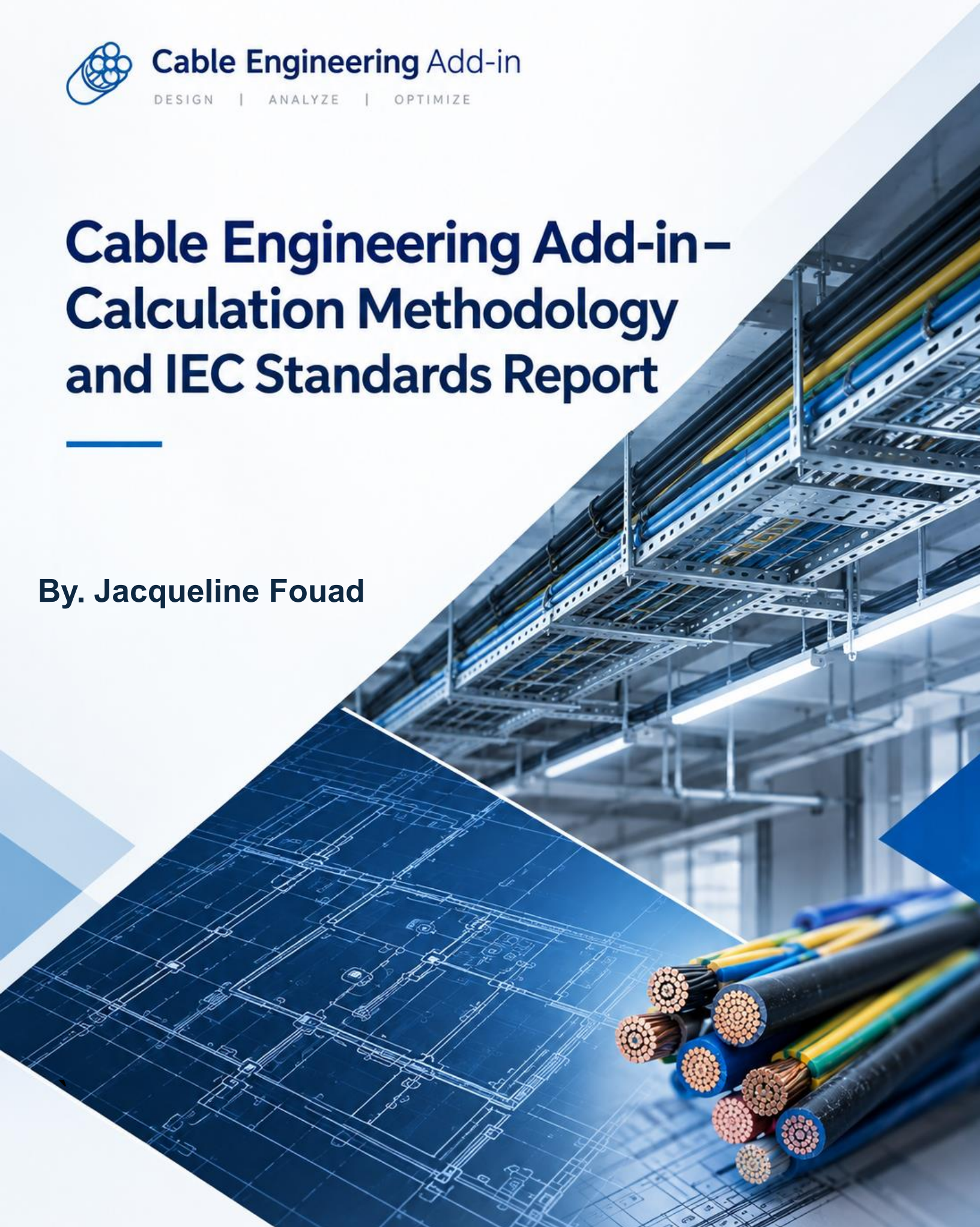
Cable Engineering Add-in

DESIGN | ANALYZE | OPTIMIZE

Cable Engineering Add-in – Calculation Methodology and IEC Standards Report

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1. Introduction

1.1 Purpose of the Report

This report documents the engineering methodologies, mathematical equations, technical parameters, and IEC reference tables used in the Cable Engineering Tool for cable sizing and electrical design calculations.

It provides a technical reference for understanding how the tool performs its calculations, how the applicable engineering requirements are incorporated into the calculation procedures, and how the resulting cable selections are evaluated against the defined acceptance criteria.

1.2 Scope of Calculation Documentation

This report covers the calculation methodologies implemented in Version 1 of the Cable Engineering Tool, including:

- Cable current-carrying capacity and the application of correction factors.
- Voltage drop calculations.
- Short-circuit withstand verification.
- Neutral conductor sizing, including harmonic current considerations.
- Protective earthing conductor sizing for TN-S and TT earthing systems.

For each calculation, the report presents the applicable equations, defines the relevant variables and technical parameters, identifies the corresponding IEC references where applicable, and describes the calculation procedure and selection criteria implemented in the tool.

The report also identifies the assumptions and limitations that affect the applicability of the documented calculation methods. It is intended to serve as a technical reference for the implemented engineering calculations and does not replace the applicable IEC standards or the separate software testing documentation.

2. Standards and Technical References

2.1 Applicable IEC Standards

2.1 Applicable IEC Standards

The following IEC standards provide the technical basis for the electrical calculation methodologies implemented in the Cable Engineering Tool.

Standard	*Title*	*Application within the Tool*
IEC 60364-5-52	Low-voltage electrical installations – Selection and erection of electrical equipment – Wiring systems	Phase cable sizing; current-carrying capacity (ampacity); application of installation, ambient temperature, and grouping correction factors; voltage drop calculations.
IEC 60949	Calculation of thermally permissible short-circuit currents, taking into account non-adiabatic heating effects	Short-circuit withstand verification of selected cables, where applicable to the implemented calculation method.
IEC 60364-4-43	Low-voltage electrical installations – Protection for safety – Protection against overcurrent	Overcurrent protection requirements relevant to cable sizing and coordination with protective devices.
IEC 60364-5-54	Low-voltage electrical installations – Selection and erection of electrical equipment – Earthing arrangements and protective conductors	Protective earthing conductor (PE) sizing for TN-S and TT earthing systems, according to the implemented calculation methods.
IEC 60364-4-41	Low-voltage electrical installations – Protection for safety – Protection against electric shock	Earthing-related protection requirements relevant to the implemented TT and TN-S verification procedures, where applicable.

2.2 Reference Tables and Clauses

This section identifies the IEC tables and clauses used as technical references for the calculation methodologies implemented in the Cable Engineering Tool. These references support the manual verification of the tool's calculations and the evaluation of the results obtained for the selected test circuits.

IEC Standard	*Reference*	*Subject*	*Related Calculation*
IEC 60364-5-52	Tables B.52.2–B.52.13, subject to the applicable installation method	Reference current-carrying capacities	Phase cable sizing
IEC 60364-5-52	Applicable grouping correction factor tables	Grouping correction factor	Cable current-carrying capacity
IEC 60364-5-52	Clause 525 and Annex G, Table G.52.1, subject to the adopted edition	Voltage drop requirements and limits	Voltage drop calculation
IEC 60364-5-52	Clause 523.6	Cable sizing where harmonic currents are present	Neutral conductor sizing
IEC 60364-4-43	Clause 431.2, with the applicable subclauses verified against the adopted edition	Protection of neutral conductors against overcurrent	Neutral conductor protection
IEC 60364-5-54	Clause 543.1.2 and Annex A	Protective conductor sizing and determination of factor k	Protective earthing conductor sizing
IEC 60364-5-54	Table 54.2	Selection of protective conductor cross-sectional area without calculating it using Clause 543.1.2	Protective earthing conductor sizing, if this method is implemented
IEC 60949	Applicable calculation provisions	Thermally permissible short-circuit current	Short-circuit withstand verification, according to the implemented method

3. Cable Current-Carrying Capacity

3.1 Calculation Methodology

The cable current-carrying capacity calculation determines whether the selected cable can safely carry the design current under the applicable installation conditions.

The calculation implemented in the Cable Engineering Tool follows these steps:

Step 1: Determine the Design Current

The design current I_b for each circuit is obtained from the Revit model.

Step 2: Determine the Circuit Breaker Rating

The circuit breaker rating I_n is selected from the applicable standard based on the installation category:

- **Residential/Commercial:** IEC 60898-1.
- **Industrial:** IEC 60947-2.

The selected rating is the next available standard rating equal to or greater than the circuit design current:

$$I_n \geq I_b$$

Step 3: Determine the Cable Derating Factors

The following correction factors are determined for each circuit according to its installation conditions:

- (k_1): Ambient temperature correction factor.
- (k_2): Grouping correction factor.
- (k_3): Installation method correction factor.

Step 4: Calculate the Required Tabulated Current-Carrying Capacity

The minimum required tabulated current-carrying capacity (I_z) is calculated using the selected circuit breaker rating and the applicable correction factors:

$$I_z = \frac{I_n}{k_1 \times k_2 \times k_3}$$

Where:

- (I_n): Selected circuit breaker rated current (A).
- (k_1), (k_2), and (k_3): Applicable cable correction factors.
- (I_z): Minimum required tabulated cable current-carrying capacity (A).

The calculated value is used as the selection threshold when evaluating the cable catalogue for suitable cable sizes.

3.2 Reference Installation Methods and Ampacity Tables

The required tabulated current-carrying capacity is calculated using the circuit breaker rating and the three correction factors described in Section 3.1.

The correction factors are determined for each circuit as follows:

- **Temperature Correction Factor (k_1):** Determined according to the applicable ambient temperature and reference temperature.
- **Grouping Correction Factor (k_2):** Determined according to the grouping conditions of the cables along their routes. The relevant factors include containment type, path arrangement, cable types installed along the path, spacing between cables, and the number of circuits sharing the path.
- **Installation Correction Factor (k_3):** Determined according to whether the circuit cable is installed in air or buried underground.

The cable properties required for these calculations are obtained from the imported Excel cable catalogue, while the installation and routing conditions are evaluated using the Revit model data.

The resulting correction factors are applied to the selected circuit breaker rating to calculate the minimum required tabulated current-carrying capacity, as described in Section 3.1.

3.3 Correction Factors (k_1, k_2, k_3)

The Cable Engineering Tool determines three correction factors for each circuit to account for the actual cable installation conditions.

3.3.1 Temperature Correction Factor (k_1)

The temperature correction factor is determined using the environmental analysis performed by the Cable Engineering Tool to account for the actual ambient temperature conditions along each cable path in the project.

The calculation follows these steps:

1. **Determine the Actual Ambient Temperature:** Obtain the ambient temperature assigned to each cable path from the tool's environmental analysis.
2. **Determine the Reference Ambient Temperature:** Identify the applicable reference temperature according to the installation condition, whether the cable is installed in air or buried underground.
3. **Retrieve the Maximum Conductor Operating Temperature:** Obtain the maximum permissible conductor temperature associated with the insulation material of the selected cable from the imported Excel cable catalogue.
4. **Calculate the Temperature Correction Factor:** Calculate (k_1) using the following equation:

$$k_1 = \sqrt{\frac{T_{max} - T_a}{T_{max} - T_r}}$$

- T_{max}Maximum permissible conductor operating temperature for the cable insulation material (°C).
- (T_a): Actual ambient temperature along the cable path (°C).
- (T_r): Reference ambient temperature applicable to the installation condition (°C).

The calculated factor is subsequently used in the current-carrying capacity calculation to account for the difference between the actual and reference ambient temperatures.

3.3.2 Grouping Correction Factor (k_2)

The grouping correction factor is determined according to the applicable grouping conditions along each cable path, following the relevant provisions and reference tables of IEC 60364-5-52.

The tool evaluates the following parameters to determine the applicable grouping factor:

1. **Containment Type:** Identifies whether the cables are installed in conduits or on cable trays.
2. **Cable Path Arrangement:** Determines whether the cables are installed touching or spaced apart.
3. **Cable Type:** Identifies whether the imported cable is single-core or multicore.
4. **Distance Between Cables:** Considers the spacing between cables along the path, where applicable.
5. **Number of Circuits:** Determines the number of circuits sharing the same cable path.

Based on these conditions, the tool selects the applicable grouping correction factor from the corresponding IEC reference table:

- **Table B.52.17:** Grouping of circuits installed in conduits in a single layer, where applicable.
- **Table B.52.20:** Grouping of multicore cables installed on cable trays, where applicable.
- **Table B.52.21:** Grouping of single-core cables installed on cable trays, where applicable.

The selected grouping correction factor is then applied in the current-carrying capacity calculation for the circuit.

The applicable table and factor depend on the installation arrangement and the conditions specified in the relevant IEC table.

3.3.3 Installation Correction Factor (k3)

The installation correction factor (k3) accounts for the effect of soil thermal resistivity on the current-carrying capacity of buried cables. The tool determines this factor according to the cable installation condition along the circuit route.

Cables installed in air:
For cable sections installed in air, the installation correction factor is set to 1.0.

Cables installed underground:
For underground cable sections, the tool uses the correction factors listed in IEC 60364-5-52, Table B.52.16, for cables installed in buried ducts. In Version 1, underground cable installation is treated as installation inside a buried duct. Direct-buried cable installation is not automatically distinguished by the tool and is outside the current implementation scope.

The soil thermal resistivity is specified by the user, who can either select one of the predefined values (1.2, 1.5, or 2.0 $K \cdot m/W$) or enter a custom value. The tool then determines the installation correction factor ($k3$) using the applicable values from IEC 60364-5-52, Table B.52.16, applying linear interpolation when the specified value falls between tabulated values and the defined boundary handling when it falls outside the tabulated range.

The correction factor is determined as follows:

- **Within the tabulated range:** If the specified soil thermal resistivity matches a tabulated value, the corresponding correction factor is used. If it falls between two tabulated values, linear interpolation is applied.
- **Below the tabulated range:** The correction factor corresponding to the minimum tabulated soil thermal resistivity is used.
- **Above the tabulated range:** The correction factor corresponding to the maximum tabulated soil thermal resistivity is used.

For linear interpolation, the correction factor is calculated using:

$$k_3 = K_{\text{lower}} + \frac{(\rho - \rho_{\text{lower}})(K_{\text{upper}} - K_{\text{lower}})}{\rho_{\text{upper}} - \rho_{\text{lower}}}$$

Where:

- (k_3): Installation correction factor.
- (ρ): Specified soil thermal resistivity.
- (ρ_{lower}): Lower tabulated soil thermal resistivity value.
- (K_{lower}): Correction factor corresponding to (ρ_{lower}).
- (ρ_{upper}): Upper tabulated soil thermal resistivity value.
- (K_{upper}): Correction factor corresponding to (ρ_{upper}).

The calculated factor is applied in the cable current-carrying capacity calculation.

Implementation limitation: The current implementation assumes buried-duct installation for underground cables. Automatic identification of direct-buried cables and selection of the corresponding correction factors are deferred to a future version, pending validation against suitable Revit project data.

3.3.4 Application of Correction Factors

The three correction factors are applied together to determine the minimum required tabulated current-carrying capacity based on the selected circuit breaker rating:

$$I_{z,\text{required}} = \frac{I_n}{k_1 \times k_2 \times k_3}$$

The resulting value is used as the minimum required cable ampacity during cable selection.

3.4 Required Current-Carrying Capacity and Cable Selection Criteria

3.4 Required Current-Carrying Capacity and Cable Selection Criteria

After calculating the voltage drop for each circuit, determining the short-circuit current and total fault-clearing time, and identifying the applicable cable material constant (k), the tool evaluates the imported cable catalogue to identify a suitable cable.

The imported cables are sorted in ascending order of conductor cross-sectional area. The tool evaluates each cable using the following short-circuit thermal withstand condition, based on the adiabatic equation:

$$I_{\text{sc}}^2 \times t \leq k^2 \times S^2$$

Where:

- (I_{sc}) Short-circuit current.
- (t): Total fault-clearing time.
- (k): Cable material constant determined according to the applicable conductor and insulation materials.
- (S): Cable conductor cross-sectional area.

The tool selects the first cable that satisfies this condition. Since the candidates are sorted by ascending cross-sectional area, the selected cable is the smallest available cable in the imported catalogue that satisfies the short-circuit thermal withstand requirement.

Minimum Required Cable Size When No Imported Cable Satisfies the Condition

If none of the imported cables satisfies the short-circuit thermal withstand condition, the tool calculates the minimum required conductor cross-sectional area by rearranging the same equation:

$$S_{\text{required}} = \frac{\sqrt{I_{\text{sc}}^2 \times t}}{k}$$

The calculated value represents the minimum theoretical conductor cross-sectional area required to satisfy the short-circuit thermal withstand condition. The tool uses this value to suggest a cable with suitable specifications, considering the applicable conductor material, insulation material, and other cable specifications available to the tool.

In Version 1, the tool suggests a cable based on the calculated minimum required cross-sectional area rather than using parallel cables to meet the short-circuit withstand requirement.

The calculation described in this section addresses short-circuit thermal withstand. The suitability of the suggested cable must also be considered in relation to the other applicable cable-sizing requirements implemented by the tool.

4.Voltage Drop Calculation

4.1 Calculation Methodology

Step 1: Determine the Design Current
The design current (I_b) for each circuit is obtained from the Revit model.

Step 2: Determine the Actual Circuit Path Length
The tool determines the circuit routing path using its Dijkstra-based routing algorithm and calculates the resulting path length. This length is used in the voltage-drop calculation.

Step 3: Import Cable Data
Cable specifications are imported from an Excel catalogue. The cable resistance and reactance are provided in ohms per kilometer $\left((\Omega/\text{km})\right)$. Before performing the voltage-drop calculation, both values are divided by 1000 to convert them to ohms per meter $\left((\Omega/\text{m})\right)$, ensuring consistency with the circuit path length expressed in metres.

$$R_{\text{m}} = \frac{R_{\text{km}}}{1000} \qquad X_{\text{m}} = \frac{X_{\text{km}}}{1000}$$

Resistance Temperature Limitation: The tool currently uses the imported cable resistance without applying a temperature correction. For accurate voltage-drop calculations, the resistance should represent the AC resistance at the applicable conductor operating temperature. If the catalogue provides resistance at 20°C only, an appropriate correction based on the conductor operating temperature and relevant AC effects is required, following the applicable IEC 60287 methodology.

Step 4: Determine the Power Factor

For mixed-load circuits, the tool uses a default power factor of 0.85 in Version 1. The corresponding sine of the phase angle is calculated as:

$$\cos\varphi = 0.85$$

$$\sin\varphi = \sqrt{1 - \cos^2\varphi}$$

The default power factor is a current implementation limitation and is intended to be refined in Version 2.

Step 5: Calculate Voltage Drop

For single-phase circuits, the voltage drop is calculated using:

$$\Delta V_{1\text{ph}} = 2I_b L(R\cos\varphi + X\sin\varphi)$$

For three-phase circuits, the voltage drop is calculated using:

$$\Delta V_{3\text{ph}} = \sqrt{3}I_b L(R\cos\varphi + X\sin\varphi)$$

Where:

- (ΔV): Calculated voltage drop in volts.
- (I_b): Circuit design current in amperes.
- (L): Actual circuit routing path length in meters.
- (R): Cable resistance per unit length, as provided by the imported catalogue.
- (X): Cable reactance per unit length, as provided by the imported catalogue.
- $(co\,\mathrm{s}\,\varphi)$: Power factor.
- $(si\,\mathrm{n}\,\varphi)$: Sine of the phase angle.

The resistance and reactance units must be consistent with the path-length unit so that the calculated voltage drop is expressed in volts.

4.2 Governing Equations

The tool calculates voltage drop using the following equations, based on the circuit phase configuration.

Single-Phase Circuits

$$\Delta V = \frac{2 I_b L}{1000} (R_{\text{km}} \cos\varphi + X_{\text{km}} \sin\varphi)$$

Three-Phase Circuits

$$\Delta V = \frac{\sqrt{3} I_b L}{1000} (R_{\text{km}} \cos\varphi + X_{\text{km}} \sin\varphi)$$

Where:

- (ΔV): Voltage drop in volts.
- (I_b): Circuit design current in amperes.
- (L): Actual circuit path length in meters, determined by the tool's routing algorithm.
- (R_{km}): Cable resistance in ohms per kilometre, obtained from the imported cable catalogue.
- (X_{km}): Cable reactance in ohms per kilometre, obtained from the imported cable catalogue.
- ($\cos\varphi$): Power factor.
- ($\sin\varphi$): Sine of the phase angle, calculated as $\left(\sqrt{1 - \cos^2 \varphi}\right)$.

For mixed-load circuits, Version 1 uses a default power factor of 0.85.

The calculated voltage drop is used to assess the voltage-drop performance of each circuit.

4.3 Electrical Parameters and Input Variables

The voltage-drop calculation uses the following electrical parameters and input variables:

Parameter	*Symbol*	*Unit*	*Source or Determination Method*
Circuit design current	(I_b)	A	Obtained from the Revit model
Actual circuit path length	(L)	m	Calculated by the tool using its Dijkstra-based routing algorithm
Cable resistance	(R)	(Ω/km)	Imported from the Excel cable catalogue
Cable reactance	(X)	(Ω/km)	Imported from the Excel cable catalogue
Power factor	($\cos\varphi$)	Dimensionless	Set to 0.85 by default for mixed-load circuits in Version 1
Sine of the phase angle	($\sin\varphi$)	Dimensionless	Calculated from the power factor using ($\sqrt{1 - \cos^2 \varphi}$)
Circuit configuration	—	—	Determines whether the single-phase or three-phase voltage-drop equation is applied

The cable resistance and reactance values imported from Excel are divided by 1000 before being used with the path length expressed in meters, converting their units from ohms per kilometer to ohms per meter.

The voltage-drop calculation uses the actual routed path length determined by the tool rather than an assumed straight-line distance.

Version 1 limitation: A fixed power factor of 0.85 is used for mixed-load circuits. A more detailed power-factor determination method is planned for a future version.

4.4 Voltage Drop Acceptance Criteria

According to IEC 60364-5-52, Annex G, Table G.52.1, the voltage-drop limits for low-voltage installations supplied directly from a public low-voltage distribution system (Type A) are:

- **Lighting circuits:** Maximum voltage drop of 3%.
- **Other loads:** Maximum voltage drop of 5%.

The tool calculates the voltage-drop percentage as follows:

$$\Delta V\% = \frac{\Delta V}{V_{\text{ref}}} \times 100$$

Where:

- (ΔV): Calculated circuit voltage drop in volts.
- (V_{ref}): Reference voltage selected according to the circuit configuration.

The reference voltage is determined as follows:

- **Single-phase circuits:** Line-to-neutral voltage.
- **Two-phase and three-phase circuits:** Line-to-line voltage.

A circuit passes the voltage-drop acceptance check when its calculated voltage-drop percentage does not exceed the applicable limit.

5.Short-Circuit Withstand Calculation

5.1 Calculation Methodology

5.1.1 Three-Phase and Two-Phase Short-Circuit Calculations

Step 1: Calculate Utility Impedance

The utility impedance is calculated using the maximum short-circuit capacity and the circuit voltage:

$$Z_Q = \frac{C_{max} \times U^2}{S_{kQ}}$$

here:

- (C_{max}) the voltage factor.
- (U) is the circuit voltage in kV.
- (S_{kQ}) is the utility short-circuit capacity in MVA.

The utility resistance and reactance are derived using the specified utility X/R ratio:

$$R_Q = \frac{Z_Q}{\sqrt{1+(X/R)^2}} \qquad X_Q = \frac{Z_Q(X/R)}{\sqrt{1+(X/R)^2}}$$

Step 2: Calculate Transformer Impedance

The transformer impedance is calculated from its percentage impedance, rated secondary voltage, rated power, and circuit voltage:

$$Z_T = \frac{U_k}{100} \times \frac{U_r^2}{S_r} \times \left(\frac{U}{U_r}\right)^2$$

The transformer resistance is calculated using the resistance voltage-drop percentage:

$$R_T = \frac{U_R}{100} \times \frac{U_r^2}{S_r} \times \left(\frac{U}{U_r}\right)^2$$

The transformer reactance is then calculated as:

$$X_T = \sqrt{Z_T^2 - R_T^2}$$

Where (U_k) is the transformer impedance voltage percentage, (U_R) is the resistance voltage-drop percentage, (U_r) is the rated secondary voltage, and (S_r) is the rated transformer power.

Step 3: Calculate Cable Resistance and Reactance

The phase-cable resistance and reactance are obtained from the imported Excel cable catalogue. Each value is multiplied by the actual routed circuit length in kilometers:

$$R_C = L_{\text{km}} \times R_{\text{catalogue}}$$

$$X_C = L_{\text{km}} \times X_{\text{catalogue}}$$

Step 4: Calculate Total Resistance and Reactance

The utility, transformer, and phase-cable contributions are added separately:

$$R_{\text{total}} = R_Q + R_C + R_T$$

$$X_{\text{total}} = X_Q + X_C + X_T$$

Step 5: Calculate Total Impedance Magnitude

$$Z_{\text{total}} = \sqrt{R_{\text{total}}^2 + X_{\text{total}}^2}2\}$$

Step 6: Calculate Maximum Short-Circuit Current

Three-phase circuits

$$I_{k3,max} = \frac{C_{max} \times U}{\sqrt{3} \times Z_{\text{total}}}$$

phase circuits

The maximum short-circuit current is calculated using the positive- and negative-sequence impedances:

$$I_{k2,max} = \frac{C_{max} \times U}{Z_1 + Z_2}$$

sequence impedances and voltage quantities must be defined consistently with the implemented two-phase fault model.

5.1.2 Single-Phase Phase-to-Neutral Short-Circuit Calculation

Step 1: Calculate Utility Resistance and Reactance

The utility impedance is calculated as:

$$Z_Q = \frac{C_{max} \times U^2}{S_{kQ}}$$

Using the specified utility X/R ratio, the resistance and reactance are derived as:

$$R_Q = \frac{Z_Q}{\sqrt{1 + (X/R)^2}} \qquad X_Q = \frac{Z_Q(X/R)}{\sqrt{1 + (X/R)^2}}$$

Step 2: Calculate Transformer Resistance and Reactance

The transformer impedance is calculated as:

$$Z_T = \frac{U_k}{100} \times \frac{U^2}{S_r}$$

The resistance voltage-drop percentage is derived from the transformer impedance voltage and the specified transformer X/R ratio:

$$U_R = \frac{U_k}{\sqrt{1 + (X/R)^2}}$$

The transformer resistance and reactance are calculated as:

$$R_T = \frac{U_R}{100} \times \frac{U^2}{S_r} \qquad X_T = \sqrt{Z_T^2 - R_T^2}$$

Step 3: Calculate Phase-Cable Resistance and Reactance

The phase-cable resistance and reactance are obtained from the imported Excel cable catalogue and multiplied by the actual routed circuit length in kilometers, using the same method as for three-phase circuits.

Step 4: Calculate Phase-to-Neutral Loop Impedance

The neutral cable is selected according to the tool's neutral-conductor sizing methodology, which considers the applicable harmonic rules.

The total resistance and reactance of the phase-to-neutral loop are calculated as:

$$R_{LN} = R_Q + R_C + R_T + R_N$$

$$X_{LN} = X_Q + X_C + X_T + X_N$$

The loop impedance magnitude is:

$$Z_{LN} = \sqrt{R_{LN}^2 + X_{LN}^2}$$

Step 5: Calculate Maximum Phase-to-Neutral Short-Circuit Current

$$I_{k,LN,max} = \frac{C_{max} \times \left(U/\sqrt{3}\right)}{Z_{LN}}$$

here (U) is the line-to-line voltage and $(U/\sqrt{3})$ is the phase-to-neutral voltage.

5.1.3 Zero-Sequence Impedance Estimation

For the phase-to-earth fault calculation, the tool estimates zero-sequence resistance and reactance using the following V1 assumptions.

Utility

$$R_{0Q} = 1.0R_Q \qquad X_{0Q} = 1.0X_Q$$

Transformer

For the current implementation, the transformer zero-sequence components are assumed to equal the corresponding transformer resistance and reactance:

$$R_{0T} = 1.0R_T \qquad X_{0T} = 1.0X_T$$

This is an implementation assumption and should be verified against the actual transformer vector group and grounding arrangement.

Cable

The current implementation applies the following multipliers to estimate the cable zero-sequence components:

$$R_{0C} = 4.0R_C \qquad X_{0C} = 3.5X_C$$

These are assumptions used by the current implementation. Their applicability depends on the cable construction and the phase and protective-earth return-path arrangement; they should not be interpreted as universal IEC-prescribed multipliers.

5.1.4 Phase-to-Earth Short-Circuit Calculation

Step 1: Calculate Positive-Sequence Resistance and Reactance

The positive-sequence resistance and reactance are calculated by summing the utility, transformer, and phase-cable contributions:

$$R_1 = R_Q + R_T + R_C \qquad X_1 = X_Q + X_T + X_C$$

Step 2: Calculate Zero-Sequence Resistance and Reactance

The zero-sequence components are estimated using the assumptions defined in Section 5.1.3:

$$R_0 = 1.0R_Q + 1.0R_T + 4.0R_C \qquad X_0 = 1.0X_Q + 1.0X_T + 3.5X_C$$

Step 3: Combine the Sequence Components

For a single-line-to-earth fault, the implementation combines two positive-sequence contributions with one zero-sequence contribution:

$$R_{LE} = 2R_1 + R_0 \qquad X_{LE} = 2X_1 + X_0$$

The total impedance magnitude is then calculated from the combined resistance and reactance:

$$Z_{LE} = \sqrt{R_{LE}^2 + X_{LE}^2}$$

This approach combines the real and imaginary components before calculating the final impedance magnitude, rather than adding the individual impedance magnitudes.

Step 4: Calculate Maximum Phase-to-Earth Short-Circuit Current

The phase-to-earth short-circuit current is calculated as:

$$I_{k,LE} = \frac{3C_{max}(U/\sqrt{3})}{Z_{LE}}$$ here $(U/\sqrt{3})$ is the phase voltage.

The accuracy of this result depends on the validity of the assumed zero-sequence components and on whether the model represents the actual earth-fault return path and system grounding arrangement.

5.1.5 Selection of Maximum Short-Circuit Current

The applicable short-circuit current values are evaluated, and the maximum calculated value is selected for the subsequent cable and protection checks.

5.2 Adiabatic Equation

The adiabatic equation is used to verify whether the selected cable conductor can withstand the thermal effects of a short-circuit current during the fault-clearing time.

The verification is based on the following equation:

$$I_{sc}^2 \times t \leq k^2 \times S^2$$

Where:

- (I_{sc}) is the short-circuit current in amperes.
- (t) is the total fault-clearing time in seconds.
- (k) is the material- and insulation-dependent constant.
- (S) is the conductor cross-sectional area in mm².

The equation can also be expressed as:

$$S_{min} = \frac{I_{sc\sqrt{t}}}{k}$$

Where (S_{min}) the minimum conductor cross-sectional area required to withstand the specified short-circuit conditions.

The selected cable is checked against the adiabatic criterion using the short-circuit current, fault-clearing time, conductor cross-sectional area, and the applicable (k) value.

The cable satisfies this thermal criterion when:

$$I_{sc}^2 t \leq k^2 S^2$$

The (k) value must correspond to the conductor material and insulation type, as well as the applicable initial and final conductor temperatures, in accordance with the relevant IEC requirements.

5.3 Cable Material Constants and Thermal Limits

The cable short-circuit thermal verification uses the conductor material factor (k) in the adiabatic equation defined in Section 5.2.

The tool stores the applicable material factors according to conductor material, insulation type, and the specified maximum insulation temperature. The values currently configured in the tool are listed below.

Table 5.3.1 — Copper Conductors

Insulation type	Maximum insulation temperature (°C)	(k) factor (A·√s/mm²)
PVC	70	115
PVC	90	100
XLPE	90	143
EPR	90	143
Rubber	60	141
Rubber	85	134
Silicone rubber	185	132

Table 5.3.2 — Aluminum Conductors

Insulation type	Maximum insulation temperature (°C)	(k) factor (A·√s/mm²)
PVC	70	76
PVC	90	66
XLPE	90	94
EPR	90	94
Rubber	60	93
Rubber	85	89
Silicone rubber	185	87

Table 5.3.3 — Steel Conductors

Insulation type	Maximum insulation temperature (°C)	(k) factor (A·√s/mm²)
PVC	70	42
PVC	90	36
XLPE	90	52
EPR	90	52
Rubber	60	51
Rubber	85	48
Silicone rubber	185	47

Application in the Tool

For each cable, the tool uses the configured material factor corresponding to the conductor material and insulation properties. The selected factor is applied in the adiabatic verification:

$$I_{sc}^2 t \leq k^2 S^2$$

The verification determines whether the selected conductor cross-sectional area can withstand the specified short-circuit current for the applicable fault-clearing time.

The configured values represent the current tool data. Their applicability must be consistent with the relevant IEC requirements and the conductor's initial and final temperature limits.

6. Neutral Conductor Sizing

6.1 Neutral Conductor Selection Methodology

Neutral conductor sizing in Cable Engineering Tool V1 has two stages. The tool first decides whether the circuit has a neutral conductor (Step 1, neutral counting). For circuits that do, it then establishes the required neutral CSA from the phase conductor CSA, conductor material, harmonic content and validation conditions, and selects a suitable cable from the imported Excel catalogue (Step 2 onward).

Step 1: Determine Whether a Neutral Conductor Is Required

Basis. The counting step applies the principle of IEC 60364 that a neutral conductor is needed only where a load, or a downstream system, is connected to it, meaning a load connected line-to-neutral. Loads connected line-to-line, and 3-wire three-phase loads such as motors, do not use a neutral. Load unbalance and harmonic content do not change whether a neutral exists. They only affect the neutral current and CSA once a neutral is present, which is handled in the sizing steps.

Decision rules.

	CIRCUIT CONDITION	NEUTRAL CONDUCTORS
1	No devices on the circuit	0
2	1-pole or 2-pole circuit feeding a panel, transformer or switchboard	1
3	1-pole or 2-pole circuit with any device at the line-to-neutral voltage (±20 V)	1
4	1-pole or 2-pole circuit, none of the above	0
5	3-pole circuit, Delta or Wye, 3-wire distribution system	0
6	3-pole circuit, Delta or Wye, 4-wire system, feeding a panel, transformer or switchboard	1
7	3-pole circuit, Delta or Wye, 4-wire system, any device flagged "Requires Neutral"	1
8	3-pole circuit, Delta or Wye, 4-wire system, any device at the line-to-neutral voltage (±20 V)	1
9	3-pole circuit, Delta or Wye, 4-wire system, none of the above (plain 3-phase loads such as motors, pumps and chillers)	0
10	Any other pole count, unrecognized system type, or wire count other than 3 or 4	1 (fallback)

Rules are evaluated in this order, and the first matching rule applies.

Implementation Note

Rows 1 to 4 and 5, 8 and 9 follow the IEC principle above. The following are **V1 design decisions, not IEC requirements**:

- IEC 60364 gives no counting rule by number of poles and no voltage tolerance. The ±20 V tolerance used to classify a device as line-to-neutral is a tool choice.
- A circuit feeding a panel, transformer or switchboard is always assigned a neutral. This is conservative. IEC requires the neutral only if the downstream system uses it, so the rule may add a neutral to feeders of panels with only line-to-line loads, or to transformers with a delta primary.
- Row 1 (empty circuit = 0) and row 10 (fallback = 1) are tool defaults. No warning is issued in V1.
- The "Requires Neutral" flag is a tool parameter. Voltage cannot distinguish a 4-wire load (such as a UPS) from a 3-wire load, because both operate at line-to-line voltage.

Known Limitations of V1

1. Equipment with a neutral terminal is detected only if its "Requires Neutral" parameter is set. Otherwise, it is treated as a plain 3-phase load.
2. A feeder to a panel is counted as 1 without checking whether the downstream panel actually serves line-to-neutral loads.

References supporting the counting principle

3. IEC 60364-1, section 312: conductor arrangements and system earthing types. A neutral is one of the possible live-conductor arrangements, and it is present only where the supply and the loads use it.
4. IEC 60364-5-52, clause 524: neutral cross-section. These rules apply only once a neutral exists (used in the sizing steps).
5. IEC 60364-4-43, clause 431: protection and disconnection of the neutral.
6. IEC 60364-5-54, clause 543.4: PEN conductor in TN-C systems.

6.2 Neutral Sizing for Single-Phase and Two-Phase Non-Pure Loads

For 1-pole and 2-pole circuits for which the neutral counting step (Section 6.1) returns one neutral conductor, the neutral cross-sectional area is set equal to the final selected phase conductor cross-sectional area:

$$S_N = S_{phase}$$

IEC 60364-5-52 (clause 524) does not permit a reduction of the neutral in single-phase and two-phase three-wire circuits, whatever the conductor size. The phase conductor CSA is first determined as the largest value required by the current-carrying capacity, voltage-drop and short-circuit withstand checks, and the neutral then takes that final value. Both conductors are counted as loaded conductors when selecting the current rating. The selected cable must exist in the imported catalogue with this phase and neutral combination.

A 2-pole circuit at line-to-line voltage has no neutral conductor (Section 6.1) and is not covered by this section.

6.3 Neutral Sizing for Three-Phase Circuits

6.3.1 Neutral Sizing for Balanced Linear Loads with Negligible Harmonic Content

This section applies to three-phase circuits for which the neutral counting step (Section 6.1) returns one neutral conductor and all connected loads are classified as linear and balanced. Circuits feeding panels, transformers, or switchboards, and circuits with no connected devices, are not classified as balanced linear circuits and are handled by the relevant sections for unbalanced or harmonic loads.

The tool supports TN-S and TT systems only; therefore, the neutral conductor is treated as a separate conductor.

Basis

IEC 60364-5-52, Clause 524.2, specifies the minimum neutral conductor cross-sectional area (CSA). For three-phase circuits, the neutral CSA must equal the line conductor CSA where the line CSA does not exceed 16 mm² for copper or 25 mm² for aluminum. Above these sizes, a reduced neutral may be used subject to the applicable requirements, including adequate current-carrying capacity and neutral conductor protection.

Step 1: Phase Conductor Selection

The phase conductor is selected first and must pass the following checks:

- Current-carrying capacity.
- Voltage drop.
- Short-circuit withstand.

The neutral conductor is then derived from the selected phase cable. The phase conductor checks are not repeated during neutral sizing.

Step 2: Minimum Neutral Cross-Sectional Area

For copper conductors:

$$S_N = S_{\text{phase}} \quad \text{when} \quad S_{\text{phase}} \leq 16\ \text{mm}^2$$

$$S_{N,min} = \max\left(0.5S_{\text{phase}}, 16\ \text{mm}^2\right) \quad \text{when} \quad S_{\text{phase}} > 16\ \text{mm}^2$$

For aluminum conductors:

$$S_N = S_{\text{phase}} \quad \text{when } S_{\text{phase}} \leq 25\ \text{mm}^2$$

$$S_{N,min} = \max\left(0.5S_{\text{phase}}, 25\ \text{mm}^2\right) \quad \text{when } S_{\text{phase}} > 25\ \text{mm}^2$$

Where:

- (S_N) is the selected neutral conductor CSA.
- $(S_{N,min})$ the minimum neutral CSA used by the V1 sizing logic.
- $\left(S_{\text{phase}}\right)$ is the selected phase conductor CSA.
- The minimum floor is 16 mm² for copper and 25 mm² for aluminum.

The candidate neutral conductor is selected from the imported cable catalogue using the same conductor material as the phase conductor. Selection starts with the smallest standard CSA not less $than$ ($S_\{N, min\}$) and proceeds upward. The selected CSA is never rounded down.

Step 3: Verification of the Reduced Neutral

Each reduced-neutral candidate is accepted only if it passes the short-circuit withstand check (3.1). The first candidate that passes is selected.

Neutral current assumption
IEC 60364-5-52 permits a reduced neutral only if the neutral current does not exceed the current-carrying capacity of the reduced neutral. For balanced linear loads, the neutral current is practically zero, so V1 assumes this condition is satisfied and does not run a separate neutral ampacity check. This is a V1 assumption, not an IEC rule.

3.1 Short-Circuit Withstand

The adiabatic condition is checked for the neutral candidate:

$$I^2t \leq k^2S_N^2$$

Where:

- (I) is the fault current used by the tool for the check.
- (t) is the applicable fault-clearing time.
- (k) is the factor appropriate to the neutral conductor's material and insulation.
- (S_N) is the candidate neutral conductor CSA.

V1 uses the same clearing time as for the phase conductor and uses the phase fault current as a conservative input for this check.

No neutral voltage-drop check is performed. The neutral current is assumed to be zero, so the neutral adds no voltage drop. The line voltage drop is governed by the phase conductors, which have already passed this check.

Step 4: Fallback

If no reduced neutral candidate passes the short-circuit check for a given phase cable, the neutral for that phase cable is set equal to the phase cable. $S_N = S_{\text{phase}}$ The phase cable has already passed all its checks

Step 5: Result

The tool returns a neutral cable list with the same length and order as the selected phase cable list. For each phase cable, the neutral entry is either the phase cable itself $S_N = S_{\text{phase}}$ or the selected reduced neutral cable. If the circuit requires no neutral (Section 6.1), the neutral list is empty. The ratio $R_N = \frac{S_N}{S_{\text{phase}}}$ is calculated from the selected sizes.

Note for the designer: *where the neutral is smaller than the phase conductor, neutral overcurrent detection that disconnects the line conductors must be provided in accordance with IEC 60364-4-43, Clause 431.*

Implementation Note

IEC-based criteria

- The 16 mm² copper and 25 mm² aluminum thresholds for equal line and neutral CSA.
- The minimum CSA floors applicable when a reduced neutral is permitted.
- The requirement that the neutral current does not exceed the capacity of the reduced neutral, and the short-circuit withstand of every live conductor.

V1 implementation decisions

- The use of 0.5 times the phase CSA as part of the minimum-size calculation.
- The tool's definitions of balanced and linear loads.
- The neutral current is assumed to be zero for balanced linear loads, so no neutral ampacity check is performed.
- No voltage-drop check is performed on the neutral.
- Where no reduced candidate passes, the neutral falls back to the phase size, per phase cable.
- The use of phase fault current as the input for the neutral short-circuit check.
- The selection of neutral candidates by conductor material and CSA from the imported catalogue.
- The use of the phase conductor clearing time for the neutral adiabatic check.

Known Limitations

- Line-to-neutral fault-loop impedance and disconnection time are not independently verified for a reduced neutral under IEC 60364-4-41.
- The neutral current is assumed to be zero for balanced linear loads. Real installations have a small residual neutral current.
- Balanced and linear classification depends on the electrical data available for connected devices in the Revit model.
- Overload protection of a reduced neutral conductor is not independently verified. Compliance with IEC 60364-4-43:2023, Clause 431.2.1, must be confirmed based on the selected protective device and its neutral protection characteristics.

6.4 Neutral Sizing for Three-Phase Balanced Non-Linear Loads

6.4.1 Harmonic Content Calculation

For balanced three-phase circuits with non-linear loads, the tool obtains the harmonic characteristics of the connected loads and calculates the total harmonic factor using the third, ninth, and fifteenth harmonic components:

$$THD_i = \sqrt{H_3^2 + H_9^2 + H_{15}^2}$$

Where:

- (H_3) is the third-harmonic component.
- (H_9) is the ninth-harmonic component.
- (H_{15}) is the fifteenth-harmonic component.

Harmonic Reduction Factors

The harmonic reduction factors implemented in Version 1 are based on IEC 60364-5-52:2009+AMD1:2024, Annex E, Table E.52.1, using the column corresponding to a neutral conductor cross-sectional area equal to the line conductor cross-sectional area.

The reduction factor is selected according to the total triple harmonic current content, (THD_{i3n}), as follows:

(THD_{i3n}) (%)	HARMONIC REDUCTION FACTOR (k_H)
5	0.99
10	0.98
15	0.96
20	0.93
25	0.89
30	0.86
35	0.82
40	0.78
45	0.74
50	0.71

The factors are stored in a configurable JSON file. When the calculated (THD_{i3n}) matches a tabulated value, the corresponding factor is selected directly. For values between two adjacent tabulated points, the tool applies linear interpolation to determine the reduction factor. This interpolation is an implementation choice and is not explicitly specified as a procedure in the IEC table.

The referenced reduction factors apply under the conditions specified in Annex E, including balanced three-phase circuits and the applicable cable and neutral-conductor arrangement.

Harmonic Reduction Factor Selection and Interpolation

When the calculated (THD_{i3n}) matches a tabulated value, the corresponding reduction factor is selected directly. When the calculated value falls between two adjacent tabulated values, the tool applies linear interpolation:

$$k_H = k_{H,1} + \frac{THD_{i3n} - T_1}{T_2 - T_1}\left(k_{H,2} - k_{H,1}\right)$$

Where:

- (k_H) is the calculated harmonic reduction factor.
- (T_1) and (T_2) are the lower and upper tabulated harmonic-content values.
- $\left(k_{H,1}\right)$ and $\left(k_{H,2}\right)$ are the corresponding reduction factors.

This interpolation is an implementation choice in V1 and is not explicitly specified as a procedure in the IEC table.

Zero Harmonic Content

When ($THD_{i3n} = 0$), the tool does not apply a harmonic reduction factor. The required phase conductor current-carrying capacity is calculated using the correction factors (k_1), (k_2), and (k_3):

$$I_{zB,\text{req}} = \frac{I_N}{k_1 k_2 k_3}$$

In this case, the required phase conductor current-carrying capacity is taken as equal to the required current-carrying capacity calculated for the neutral conductor, (I_N), before applying the installation correction factors.

6.4.2 Required Current-Carrying Capacity

The minimum required cable current-carrying capacity is calculated by applying the harmonic reduction factor together with the applicable installation correction factors:

$$I_{zB,\text{req}} = \frac{I_{\text{design}}}{k_1 \times k_2 \times k_3 \times k_H}$$

Where:

- ($I_{zB,\text{req}}$) is the minimum required reference current-carrying capacity of the cable before applying the correction factors.
- (I_{design}) is the circuit design current used by the tool.
- (k_1) is the ambient-temperature correction factor.
- (k_2) is the grouping correction factor.
- (k_3) is the applicable additional correction factor.
- (k_H) is the harmonic reduction factor selected from Table E.52.1 or obtained by linear interpolation between adjacent tabulated values.

The harmonic reduction factor decreases the allowable current-carrying capacity of the cable. Therefore, the required reference current-carrying capacity increases as the reduction factor decreases. The separate multiplier previously used for (K_H=1.45) is no longer applied under this method.

Reference: IEC 60364-5-52:2009+AMD1:2024, Annex E, Table E.52.1.

6.4.3 Neutral Cable Selection

The default neutral CSA is initially set equal to the phase CSA. The subsequent selection procedure depends on the calculated $THDi$.

Neutral Short-Circuit and Voltage-Drop Checks

For each neutral cable candidate, the tool performs short-circuit withstand and voltage-drop checks in addition to the current-carrying capacity check.

The neutral voltage-drop calculation uses the design neutral current determined for balanced non-linear loads. In Version 1, this current is calculated from the third-harmonic component as follows:

$$I_{N,\text{design}} = 3 \times I_{phase,A} \times H_3$$

Where:

- $(I_{N,\text{design}})$ is the design neutral current used for the voltage-drop calculation.
- $(I_{\text{phase,A}})$ is the phase A current in amperes.
- (H_3) is the third-harmonic current component expressed as a decimal ratio.

The candidate neutral cable is accepted only when it satisfies the applicable current-carrying capacity, short-circuit withstand, and voltage-drop conditions.

Case A — $THDi$ less than or equal to 15%

1. If the phase conductor is copper with a CSA of 16 mm² or less, or aluminum with a CSA of 25 mm² or less, the initial neutral CSA is set equal to the phase CSA.
2. For larger phase conductors, the minimum neutral CSA is calculated using the rule defined in Section 6.3.1.
3. The tool determines the required current-carrying capacity and evaluates candidate neutral cables from the imported catalogue.
4. The phase-equivalent cable is retained if it satisfies the required current-carrying capacity, short-circuit withstand, and voltage-drop conditions.
5. If the phase-equivalent cable fails any of these checks, the tool searches the imported catalogue and selects the first candidate satisfying all applicable conditions.

Case B — $THDi$ greater than 15%

The tool searches the imported cable catalogue for a suitable neutral conductor using the calculated harmonic-adjusted current-carrying capacity. Each candidate is evaluated against the following conditions:

- Corrected current-carrying capacity meets or exceeds the required capacity.
- Short-circuit withstand condition is satisfied.
- Voltage-drop condition is satisfied.

The first candidate satisfying all applicable conditions is selected.

For both cases, cable availability is limited to the imported Excel catalogue and its available conductor sizes and specifications.

6.5 Neutral Sizing for Three-Phase Unbalanced Linear Loads

For three-phase circuits with unbalanced pure linear loads, the tool first checks the phase conductor material and CSA.

If the phase conductor is copper with a CSA of 16 mm² or less, or aluminum with a CSA of 25 mm² or less, the neutral CSA is set equal to the phase CSA.

For larger phase conductors, the tool calculates the fundamental neutral current from the phase currents:

$$I_{N,\text{fund}} = \sqrt{I_A^2 + I_B^2 + I_C^2 - I_A I_B - I_B I_C - I_C I_A}$$

Where (I_A), (I_B), and (I_C) are the phase currents used by the tool.

For each candidate neutral cable, the corrected current-carrying capacity is calculated as:

$$I_{z,\text{corrected}} = I_z, k_1, k_2, k_3$$

The candidate must satisfy:

$$I_{z,\text{corrected}} \geq I_{N,\text{fund}}$$

The tool then checks the candidate against the short-circuit withstand condition and the applicable voltage-drop criterion. The first cable in the imported catalogue satisfying all these conditions is selected.

6.6 Neutral Sizing for Three-Phase Unbalanced Non-Linear Loads

6.6.1 Harmonic Content Calculation

For three-phase unbalanced non-linear loads, the tool calculates the combined harmonic content using the third, ninth, and fifteenth harmonic values:

$$THD_{i,3n} = \sqrt{H_3^2 + H_9^2 + H_{15}^2}$$

where (H_3), (H_9), and (H_{15}) represent the third, ninth, and fifteenth harmonic percentages, respectively.
The calculated harmonic content determines the neutral conductor sizing procedure.

6.6.2 Neutral Conductor Selection Criteria

Case 1: Harmonic content below 15%

When the calculated harmonic content is below 15%, the tool selects a neutral conductor with the same cross-sectional area (CSA) as the selected phase conductor, provided that the phase cable has passed the voltage-drop and adiabatic short-circuit checks.

Case 2: Harmonic content from 15% to 33%

When the calculated harmonic content is greater than or equal to 15% and less than or equal to 33%, the target neutral current is calculated as:

$$I_{N,target} = \frac{I_{design}}{0.86}$$

The tool searches the imported cable catalogue for candidate cables whose CSA is greater than or equal to the selected phase conductor CSA.
The corrected current-carrying capacity of each candidate is calculated as:

$$I_{z,corrected} = I_z \times k_1 \times k_2 \times k_3$$

Candidates whose corrected current-carrying capacity is lower than the target neutral current are rejected. The remaining candidates proceed to the voltage-drop and adiabatic short-circuit checks.

Case 3: Harmonic content greater than 33% and below 45%

When the calculated harmonic content is greater than 33% and below 45%, the target neutral current is calculated as:

$$I_{N,target} = \frac{I_N}{0.86}$$

The tool evaluates candidate neutral cables against the target current and CSA requirements. Candidates must also satisfy the voltage-drop and adiabatic short-circuit criteria.

Case 4: Harmonic content of 45% or higher

When the calculated harmonic content is greater than or equal to 45%, the target neutral current is calculated as:

$$I_{N,target} = \frac{I_N}{1.0}$$

The tool evaluates candidate cables against the target current and CSA requirements. The selected cable must also pass the voltage-drop and adiabatic short-circuit checks.

6.6.3 Neutral Current Calculation

For the voltage-drop assessment, the tool estimates the fundamental neutral current from the three phase-current values:

$$I_{N,fund} = \sqrt{I_A^2 + I_B^2 + I_C^2 - I_A I_B - I_B I_C - I_C I_A}$$

The third, ninth, and fifteenth harmonic neutral-current contributions are calculated as follows:

$$I_{N,h3} = (I_A + I_B + I_C)H_3$$

$$I_{N,h9} = (I_A + I_B + I_C)H_9$$

$$I_{N,h15} = (I_A + I_B + I_C)H_{15}$$

The combined harmonic neutral current is calculated as:

$$I_{N,h} = \sqrt{I_{N,h3}^2 + I_{N,h9}^2 + I_{N,h15}^2}$$

The total estimated neutral current is then:

$$I_N = \sqrt{I_{N,fund}^2 + I_{N,h}^2}$$

The calculated neutral current is used for the voltage-drop assessment. The selected neutral cable must also satisfy the adiabatic short-circuit criterion.

6.6.4 IEC Reference

The reduction factors of 0.86 and 1.0 and the associated harmonic-content thresholds are referenced to IEC 60364-5-52:2009+AMD1:2024 CSV, Edition 3.1, Table E.52.1.

The combined harmonic-content calculation and its application to three-phase unbalanced non-linear loads describe the tool's V1 implementation. Their applicability to unbalanced systems should be treated as an implementation-specific engineering assumption rather than direct confirmation of IEC compliance.

6.7 Final Neutral Cable Validation and Selection

After establishing the applicable neutral sizing requirements, the tool evaluates candidate cables from the imported Excel catalogue.

6.7.1 Current-Carrying Capacity

For the applicable calculation method, the candidate cable must satisfy the required current-carrying capacity after the relevant correction factors and harmonic treatment have been applied.

6.7.2 Short-Circuit Withstand

The tool evaluates the adiabatic short-circuit condition:

$$I_{sc}^{2}, t \leq k^{2}S^{2}$$

Where:

- (I_{sc}) is the short-circuit current used in the calculation.
- (t) is the total clearing time.
- (k) is the cable material and insulation constant selected by the tool.
- (S) is the candidate conductor CSA.

6.7.3 Voltage Drop

The candidate neutral cable is also checked against the voltage-drop criterion implemented in the tool.

6.7.4 Selection Result

A candidate is accepted only when it satisfies the applicable current-carrying-capacity, short-circuit, and voltage-drop checks. Where the initial phase-equivalent neutral cable fails, the tool searches the imported catalogue and selects the first candidate that passes the required checks.

If no available candidate satisfies the conditions, the tool cannot identify a suitable neutral cable from the imported catalogue under the implemented selection procedure.

6.8 Version 1 Scope and Limitations

The neutral-sizing functionality in Cable Engineering Tool V1 combines conductor-size rules, harmonic-related capacity adjustments, neutral-current calculations, and cable validation checks.

The following limitations should be considered when interpreting its results:

1. The harmonic factors 0.96, 0.84, 0.85, and 1.45 are fixed V1 values. Their use does not establish that they reproduce the applicable IEC method for every installation arrangement or harmonic-current range.
2. The harmonic calculation includes the third, ninth, and fifteenth harmonic components. It should not be assumed to represent every possible harmonic component or every IEC definition of harmonic content.
3. The fallback of using 50% of the phase CSA when the required neutral size is absent from the catalogue is an implementation assumption, not a general IEC rule.
4. For three-phase unbalanced non-linear loads, V1 follows the fundamental neutral-current calculation used for unbalanced linear loads and does not separately quantify the harmonic contribution to neutral current.
5. Candidate selection depends on the cable sizes and technical data available in the imported Excel catalogue.
6. The implemented rules and factors require verification against the applicable IEC edition and the installation conditions before making an unrestricted claim of IEC compliance.

This section documents the neutral-sizing methodology implemented in V1. It distinguishes the tool's current calculation procedure from requirements that may need further verification against the applicable IEC standard.

7.Protective Earthing Conductor Sizing

7.1 General Methodology

The tool supports protective earthing conductor sizing for TN-S and TT earthing systems only. The user can select the earthing system to be applied to the project's circuits, and the tool performs the earthing conductor sizing calculations according to the selected system.

7.2 TN-S Earthing System Calculation

The tool determines the recommended protective earthing conductor cross-sectional area (CSA) for the TN-S earthing system using two methods:

1. **Equation-Based Method:** Calculates the required earthing conductor CSA using the prospective short-circuit fault current and conductor material factor, then checks the result against the minimum CSA requirements.
2. **IEC Table-Based Method:** Determines the recommended earthing conductor CSA using the configured IEC 60364 table data.

The tool compares the results of both methods and selects the **larger CSA** as the final recommended earthing conductor size.

7.2.1 Equation-Based Method

7.2.1.1 Prospective Short-Circuit Fault Current Calculation

The tool calculates the prospective short-circuit fault current using the utility, transformer, and cable impedance components.

Utility impedance

The utility impedance is calculated from the utility short-circuit capacity:

$$Z_{utility} = \frac{C_{max} \times U^2}{S_{sc}}$$

where:

- $(C_\{max\})$ is the voltage factor.
- (U) is the circuit voltage in kV.
- (S_{sc}) is the utility short-circuit capacity in MVA.

The utility resistance and reactance are calculated using the utility reactance-to-resistance ratio:

$$R_{utility} = \frac{Z_{utility}}{\sqrt{1 + r_{utility}^2}}$$

$$X_{utility} = \frac{Z_{utility} \times r_{utility}}{\sqrt{1 + r_{utility}^2}}$$

where $(r_{utility})$ is the utility reactance-to-resistance ratio.

Transformer impedance

The transformer impedance is calculated using the transformer impedance voltage, rated power, and circuit voltage:

$$Z_{transformer} = \frac{U_{imp}}{100} \times \frac{U^2}{S_{rated}}$$

The resistance voltage-drop percentage is calculated as:

$$U_R = \frac{U_{imp}}{\sqrt{1 + r_{transformer}^2}}$$

The transformer resistance and reactance are then calculated as:

$$R_{transformer} = \frac{U_R}{100} \times \frac{U^2}{S_{rated}}$$

$$X_{transformer} = \sqrt{Z_{transformer}^2 - R_{transformer}^2}$$

where:

- (U_{imp}) is the transformer impedance voltage percentage.
- (S_{rated}) is the transformer rated power in MVA.
- $(r_{transformer})$ is the transformer reactance-to-resistance ratio.
- $(U)$is the circuit voltage in kV.

Cable impedance

The cable resistance and reactance are retrieved from the imported Excel cable catalogue.

The total loop resistance and reactance are calculated by adding the utility, transformer, and cable contributions:

$$R_{loop} = R_{utility} + R_{transformer} + R_{cable}$$

$$X_{loop} = X_{utility} + X_{transformer} + X_{cable}$$

The tool also calculates the zero-sequence resistance and reactance using the following expressions:

$$R_0 = R_{utility} + R_{transformer} + 3R_{cable}$$

$$X_0 = X_{utility} + X_{transformer} + 3X_{cable}$$

The total loop impedance is then calculated as:

$$Z_{loop} = \sqrt{R_{loop}^2 + X_{loop}^2}$$

Finally, the prospective short-circuit fault current is calculated as:

$$I_{sc} = \frac{\sqrt{3} \times C_{max} \times U_{LL} \times 1000}{Z_{loop}}$$

where $(U_{LL})$is the line-to-line voltage in kV, and $\left(Z_{loop}\right)$ is the total loop impedance in ohms.

7.2.1.2 Conductor Material Factor

The tool retrieves the conductor material factor (k) according to the conductor material and insulation type and temperature rating.

The configured values are based on the following material and insulation combinations:

CONDUCTOR MATERIAL	INSULATION	MAXIMUM INSULATION TEMPERATURE	FACTOR (K)
COPPER	PVC	70°C	115
COPPER	PVC	90°C	100
COPPER	XLPE	90°C	143
COPPER	EPR	90°C	143
COPPER	Rubber	60°C	141
COPPER	Rubber	85°C	134
COPPER	Silicone rubber	185°C	132
ALUMINUM	PVC	70°C	76
ALUMINUM	PVC	90°C	66
ALUMINUM	XLPE	90°C	94
ALUMINUM	EPR	90°C	94
ALUMINUM	Rubber	60°C	93
ALUMINUM	Rubber	85°C	89
ALUMINUM	Silicone rubber	185°C	87
STEEL	PVC	70°C	42
STEEL	PVC	90°C	36
STEEL	XLPE	90°C	52
STEEL	EPR	90°C	52
STEEL	Rubber	60°C	51
STEEL	Rubber	85°C	48
STEEL	Silicone rubber	185°C	47

7.2.1.3 Required Earthing Conductor CSA

The required earthing conductor CSA is calculated using the adiabatic equation:

$$S_{required} = \frac{\sqrt{I_{sc}^2 \times t}}{k}$$

where:

- $(S_{required})$ is the required earthing conductor CSA in mm².
- (I_{sc}) is the prospective short-circuit fault current in amperes.
- (t) is the operating time in seconds.
- (k) is the conductor material factor selected according to the conductor material and insulation characteristics.

7.2.1.4 Minimum CSA Requirement

The tool also checks the calculated CSA against the configured minimum earthing conductor sizing table.

For earthing conductors made of the same material as the associated line conductor, the minimum CSA is determined as follows:

Line conductor CSA (S)	Minimum earthing conductor CSA
$(S \leq 16\ \text{mm}^2)$	(S)
$(16 < S \leq 35\ \text{mm}^2)$	$(16\ \text{mm}^2)$
$(S > 35\ \text{mm}^2)$	$(S/2)$

For earthing conductors made of a different material, the corresponding minimum CSA is adjusted using the material factors:

$$S_{PE,min} = \frac{k_1}{k_2} \times S_{PE,reference}$$

where (k_1) and (k_2) represent the relevant material factors, and $\left(S_{PE,reference}\right)$ is the minimum CSA obtained from the corresponding sizing range.

The tool compares the adiabatic calculation result with the minimum CSA requirement and retains the larger value as the recommendation from the equation-based method.

7.2.2 IEC Table-Based Method

In the second method, the tool determines the recommended protective earthing conductor CSA using the configured `TableIEC60364` data.

The selected value is obtained from the applicable table entry based on the available circuit and conductor sizing information. This method provides an independent table-based recommendation for comparison with the equation-based result.

7.2.3 Final Earthing Conductor Selection

After completing both methods, the tool compares their recommended CSA values:

$$S_{PE,final} = \max\left(S_{PE,equation}, S_{PE,table}\right)$$

The larger CSA is selected as the final recommended protective earthing conductor size for the TN-S earthing system.

7.3 TT Earthing System Calculation

The tool determines the recommended protective earthing conductor cross-sectional area (CSA) for TT earthing systems using the configured sizing rules based on IEC 60364-5-54.

7.3.1 Standard CSA Selection

The tool selects the standard protective earthing conductor CSA according to the phase conductor CSA:

Phase conductor CSA	Standard earthing conductor CSA
$(S \leq 16\ \text{mm}^2)$	(S)
$(16 < S \leq 35\ \text{mm}^2)$	$(16\ \text{mm}^2)$
$(S > 35\ \text{mm}^2)$	$(S/2)$

7.3.2 Buried Cable Adjustment

The tool identifies a circuit cable as buried when its K3 factor is less than 1. In the current V1 implementation, cable protection status is not evaluated independently; the tool assumes that the cable is unprotected.

When the cable is identified as both buried and unprotected, the tool applies an additional minimum CSA requirement. If the standard earthing conductor CSA calculated above is less than $(16\ \text{mm}^2)$, the recommended CSA is increased to $(16\ \text{mm}^2)$.

Otherwise, the standard CSA is retained.

7.3.3 Final CSA Selection

The final recommended TT protective earthing conductor CSA is determined by applying the buried and unprotected cable adjustment, where applicable, to the standard CSA obtained from the sizing table.

7.4 Earthing Conductor Selection Criteria

After determining the recommended protective earthing conductor CSA according to the selected earthing system, the tool searches the imported cable catalogue for a suitable cable.

The selection process considers the following criteria:

1. **Cross-Sectional Area (CSA):** The selected cable must satisfy the recommended minimum earthing conductor CSA calculated for the selected earthing system.
2. **Cable Availability:** The selected size must be available in the imported cable catalogue.
3. F**inal Selection:** The tool selects a suitable cable based on the calculated CSA requirement and the available catalogue sizes.